

An Accurate and Efficient Zero-Crossing Line Classifier for Multiscale Parameter Estimation of Gaussian Signals Subject to Noise

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Abstract—The multiscale parameter estimation framework is a method for estimating the true parameters of signals subject to noise. The method is based on detecting lines of zero-crossings within the Continuous Wavelet Transform and substituting their locations in time into analytical equations directly expressing the unknown signal parameters. Evidently, this approach depends on selecting the correct lines, corresponding to the signal of interest and not to other phenomena related to noise. This task can be posed as the binary classification problem of determining for each zero-crossing line found whether or not it should be used for parameter estimation. It has been shown that even for very high noise levels, a correct classification leads to very accurate estimates, while a wrong classification results in highly inaccurate estimates. Therefore, with this particular approach the classification of zero-crossing lines poses the limiting factor to the accuracy of the estimated parameters. In this work, we propose a novel, efficient and more robust classifier called "stencil operator" which accurately detects the best combination of zero-crossing lines of Gaussian input signals. We evaluate the performance of this new classifier using synthetic Gaussian signals subject to white (Gaussian) noise with signal-to-noise ratios ranging from 50 dB to −20 dB. By studying the error between estimated and ground truth parameters, we show that the new classifier outperforms the current method for all noise levels considered and for a noise level of e.g. −12 dB improves the median error from 132% to 28%. The proposed classifier pushes the boundary for analyzing heavily disturbed signals using multiscale parameter estimation to a new level.

Index Terms—Continuous Wavelet Transform, Time-Scale analysis, Gaussian functions, Zero-crossings, Parameter Estimation, Signal-to-noise ratio

I. INTRODUCTION

Detecting and characterizing a bump or peak in noisy data is one of the most fundamental tasks in signal processing applica-

tions. In the past, a multitude of different algorithms have been proposed, ranging from simple time-series analysis to complex machine learning models. Typical fields of application are the detection and characterization of peaks in biosignals [1], mass spectrometry [2], or climate data [3].

The method proposed in this work extends the recently introduced multiscale parameter estimation (msPE)¹ framework [4] which can be used for measuring the true parameters of a peak subject to noise from its continuous wavelet transform (CWT) coefficients. The msPE framework allows for estimating the parameters of peaks following a generalization of the 2nd Gaussian derivative of which the Gaussian function is a case of. In previous work [4], we demonstrated that msPE outperforms state-of-the-art methods for estimating the parameters of a Gaussian function subject to very high levels of noise [5]–[7].

The msPE is based on detecting lines of zero-crossing within the CWT and for each line a binary classification is made to decide whether the line is used for parameter estimation or not. If only zero-crossing lines stemming from the underlying model function, and not from noise, are used, this method allows to estimate the parameters accurately. However, for msPE this poses a limiting factor: If one or multiple lines originating from noise are used for parameter estimation, results are entirely wrong, greatly limiting the potential of msPE.

Therefore, in this work we propose a "stencil operator" which constitutes an efficient method for classifying zero-crossing lines and demonstrates its positive influence on the

¹An open-source implementation in MATLAB programming language can be found at <https://github.com/nspi/msPE>

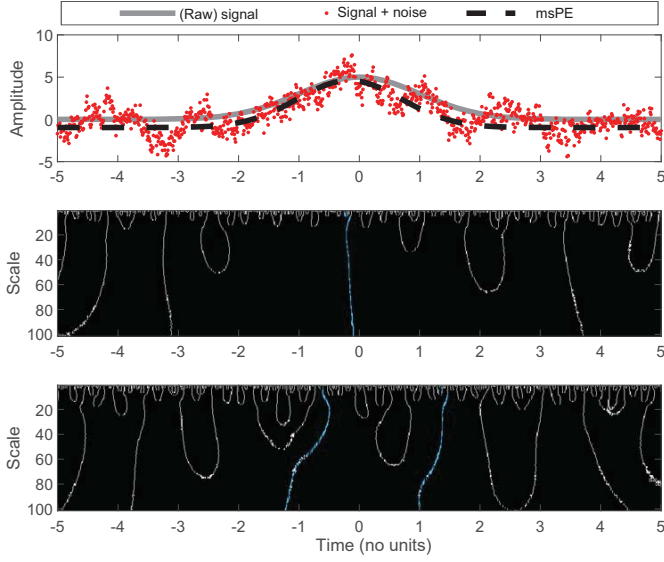


Fig. 1. *Top*: Gaussian signal (gray curve) with additive correlated pink noise $f(t)$ (red dots) and signal estimated by msPE (black curve). *Center*: Zero-crossings in WT^1 . *Bottom*: Zero-crossings in WT^2 . Lines of zero-crossings used for msPE are displayed in blue. As can be seen, the estimated parameters are similar to the true values.

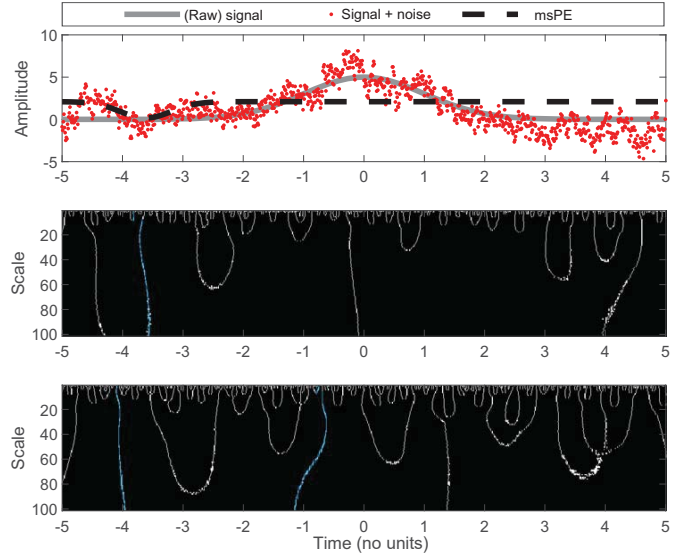


Fig. 2. *Top*: Gaussian signal (gray curve) with additive pink noise $f(t)$ (red dots) and signal estimated by msPE (black curve). *Center*: Zero-crossings in WT^1 . *Bottom*: Zero-crossings in WT^2 . Lines of zero-crossings used for msPE are displayed in blue. As can be seen, estimated parameters are entirely wrong due to a wrong line classification.

overall accuracy of msPE.

In section II, we introduce the fundamentals of msPE, the principle of the proposed "stencil operator" and the evaluation methodology. Section III presents the results which are discussed in section IV. In the final section V we discuss pros and cons of the proposed method and outline avenues for future work.

II. MATERIAL AND METHODS

A. Continuous Wavelet Transform

A one-dimensional signal $f(t)$ can be represented by the CWT at scale s and point in time u :

$$WTf(u, s) = \int_{-\infty}^{\infty} f(t)\psi(t) dt \quad (1)$$

where $\psi(t)$ is a real-valued function:

$$\psi(t) = \frac{1}{\sqrt{s}} \Psi\left(\frac{t-u}{s}\right) \quad (2)$$

derived from a "mother wavelet" $\int_{-\infty}^{\infty} \Psi(t) dt = 0$. Due to the scale factor s and its effect on the wavelet, the CWT enables the analysis of $f(t)$ with a flexible time-frequency resolution. This has certain advantages over time-frequency analysis methods with a fixed time-frequency resolution, e.g., the short-time Fourier Transform. Mallat and Hwang [8] have shown that – if a Gaussian wavelet $\theta_s(t) = 1/\sqrt{s} \exp(-t^2/s^2)$ is applied – eq. (1) can be written as:

$$WT^n f(u, s) = s^n \frac{d^n}{du} (f \star \theta_s)(u) \quad (3)$$

with $\star$ denoting convolution. This multiscale differential operator represents the CWT as the n^{th} derivative of $f(t)$ smoothed

by θ_s . This smoothing acts as a low-pass filter with a stopband that increases with s . In the following we use the first-order derivative (WT^1) and second-order derivative (WT^2).

B. Multiscale parameter estimation

In previous work [4] we developed the msPE framework for estimating the parameters of an input signal $f(t)$ from eq. (3). The msPE is based on the idea of making use of the low-pass filtering effect of the CWT by deriving scale-dependent algebraic equations for each parameter of a model function following a generalization of the second-order Gaussian derivative of which the Gaussian function is a case of. In this work, we limit our analysis to input signals best represented by the Gaussian function:

$$G_{A,\sigma,\mu,B}(t) = A \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right) + B. \quad (4)$$

Parameter A represents the amplitude, σ the width, μ the location, and B the constant baseline of input signal $f(t)$.

In [4] we derived closed-form solutions for all four parameters from zero-crossings of WT^1 and WT^2 denoted by $u0^1$ and $u0^2$, respectively. Fig. 1 and Fig. 2 show Gaussians subject to noise and the corresponding zero-crossings $u0^1$ and $u0^2$ forming lines in WT^1 and WT^2 , respectively. A Gaussian input function $f(t)$ results in zero-crossings at each scale of WT^1 representing the maximum of $f(t)$ and two zero-crossings in WT^2 representing the inflection points of $f(t)$. Although the noise also affects the zero-crossings, this effect decreases with increasing scale due to the filtering effect (convolution) of the CWT and therefore zero-crossings at high scales can be used for accurate parameter estimation.

Parameter μ describing the center of $f(t)$ can be derived directly from the central WT^1 zero-crossing denoted $u0^1$:

$$\mu(s) = u0^1(s) \quad (5)$$

while the computation of the width requires a WT^2 zero-crossing:

$$\sigma(s) = \pm \sqrt{(u0^2(s) - \mu(s))^2 - \frac{s^2}{2}}. \quad (6)$$

Parameters A and B cannot be computed from the CWT and we provide solutions based on the convolution:

$$B(s) = \frac{1}{\sqrt{\pi}\sqrt{s}} \left(\frac{f_c(\mu, s) - f_c(\mu - \sigma, s)}{\exp(K\sigma^2) - 1} - f_c(\mu - \sigma, s) \right) \quad (7)$$

$$A(s) = -\frac{\sqrt{\frac{2\sigma^2}{s}} + s \left(\sqrt{\pi} B \sqrt{s} + f_c(\mu, s) \right)}{\sqrt{2\pi}\sigma} \quad (8)$$

with $f_c(u, s) = (G_{A,\sigma,\mu,B} \star \theta_s)(u)$ and $K = 1/(s^2 + 2\sigma^2)$.

In summary, msPE allows for estimating all model parameters of a Gaussian input signal (4) (amplitude A , width σ , location μ , baseline B) by substituting zero-crossings of WT^1 and WT^2 into equations (5)-(8).

C. The problem of zero-crossing line classification

A limiting factor for the accuracy of msPE is given by the step of heuristically determining those zero-crossings associated with the input signal. A useful strategy is to detect lines of zero-crossings as it has been proven that those are never interrupted if eq. (3) is used [9]. Fig. 1 shows an example for a signal disturbed by correlated noise. As can be seen, the correct lines were chosen, resulting in fairly accurate parameters, i.e. a large agreement between the estimated and true waveform. But, at high noise levels this is not a trivial task as shown in Fig. 2. Therefore, at the core of msPE lies the fundamental problem of classifying each zero-crossing line.

In our previous work [4] we followed a basic approach called "variance operator" and started by detecting the one WT^1 zero-crossing line with lowest variance, e.g. the straightest line and regarding it as the central WT^1 line associated with parameter μ . As we know that a Gaussian input signal results in two WT^2 lines on the left and right of the central WT^1 line (Fig. 1), we picked the nearest WT^2 lines. Despite using this simple heuristic it outperformed state-of-the-art methods for Gaussian parameter estimation [5]–[7]. However, simply looking for the straightest line is prone to error and a more sophisticated method is needed to improve robustness.

D. Zero-crossing line classification via stencil operator

Here we propose the "stencil operator" as a new line classifier for msPE. As shown in Fig. 3, the maximum of a Gaussian function $f(t)$ corresponds to a zero-crossing of the first-order derivative (WT^1) and additionally to a global extremum of the second-order derivative (WT^2). While the "variance operator" only considers zero-crossings of WT^1 , the above observation suggests to incorporate the raw CWT

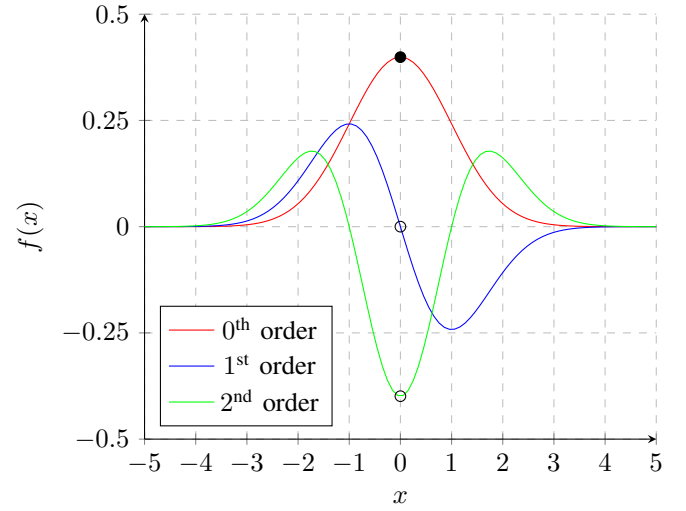


Fig. 3. Plots of the Gaussian derivative function for order 0 (red), 1 (blue), and 2 (green). The filled circle represents the peak of the Gaussian function. This peak corresponds to the unfilled circles within the 1st and 2nd order derivatives.

coefficients of WT^2 as well. The basic idea is to choose the one WT^1 zero-crossing line of the highest absolute mean value over all scales of WT^2 coefficients at the same position in time. The name "stencil operator" originates from the idea of placing a stencil of WT^1 lines over the WT^2 coefficients.

The "stencil operator" is computed in three steps. First, for each WT^1 zero-crossing line the related WT^2 coefficients are read. In the second step, for each WT^1 line the mean value over all scales of the "stenciled" absolute WT^2 coefficients is computed. Finally, the one WT^1 line of highest mean value is classified as being the central line corresponding to parameter μ . Furthermore, the nearest WT^2 lines on its left and right are chosen.

E. Experiment

To evaluate the new operator's ability to classify the central zero-crossing lines within noisy signals, we integrate the "stencil operator" into the msPE framework and compare it directly to the "variance operator" proposed in [4].

The experiment is based on synthetic signals and we investigate our method's accuracy as a function of SNR. We start by defining our discrete signal and noise model:

$$\tilde{f}[t_i] = f[t_i] + \eta[t_i] \quad \text{with } t_i \in [t_{\min}, t_{\max}], \quad (9)$$

where $f[t_i]$ is a Gaussian (eq. (4)) with additive normally-distributed noise $\eta[t_i]$. The resulting signal $\tilde{f}[t_i]$ is analyzed by msPE. We compute the standard deviation σ_η to obtain a constant SNR across varying model parameters:

$$\text{SNR [dB]} := 10 \log_{10} \left(\frac{\text{Var}(f[t_i])}{\text{Var}(\eta[t_i])} \right) \quad \text{and } \sigma_\eta = \sqrt{\frac{\text{Var}(f[t_i])}{10^{\frac{\text{SNR}}{10}}}}$$

with $t_i \in [t_{\min}^{\text{SNR}}, t_{\max}^{\text{SNR}}]$, (10)

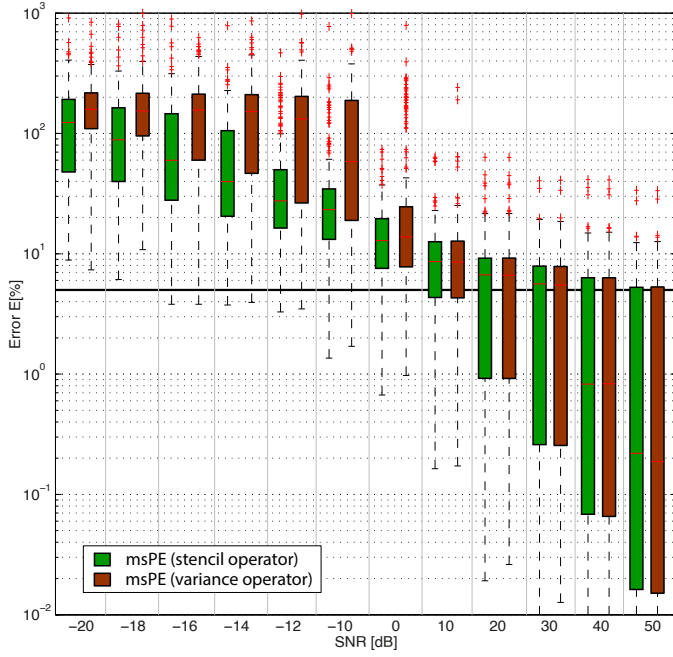


Fig. 4. Log-log graph, showing error distribution using boxplots: All parameters of our 3-param. model, corrupted by normally-distributed noise of varying SNR were estimated using two different zero-crossing line classification operators. Each boxplot (whiskers at $1.5 \times IQR$) represents the distribution of the relative error E [%] of 256 signals estimations with varying model parameters and constant SNR.

where $\text{Var}()$ is the variance operator. It is ensured that the SNR only reflects the peak of the Gaussian and not the flat part of the signal by letting $t_{min}^{SNR} = \mu - 3\sigma$ and $t_{max}^{SNR} = \mu + 3\sigma$.

We define 12 SNR levels ranging from -20 to 50 dB in increments of 2 dB below -10 dB and increments of 10 dB above. For each SNR, a set of 256 signals is generated according to our model (eq. (4)).

Parameter values are obtained by allowing parameters A , μ and B to assume values from the set $\{-2, -1, 1, 2\}$, respectively. Additionally, parameter σ can assume values from the set $\{0.5, 1, 1.5, 2\}$. This results in a total of $4^4 = 256$ signals. Each signal $f(t)$ is sampled at a rate of 0.01 Hz to obtain $f[t_i]$, with $t_i \in [-15, 15]$ ($N = 3000$ samples) and Gaussian noise is added according to eq. (9).

We apply msPE with the "stencil operator" and msPE with "variance operator" to all synthetic signals and analysed scales $[1, 150]$. At the end of the signals interval $[-15, 15]$ false zero-crossings might be introduced. Therefore, only the interval $[-12, 12]$ is considered by msPE.

For each run of msPE an error metric E is computed:

$$E[\%] := \frac{\|\mathbf{P}_n - \tilde{\mathbf{P}}_n\|_2}{\|\mathbf{P}_n\|_2} \times 100. \quad (11)$$

$P = (A, \sigma, \mu, B)^T$ is the vector representing the true model parameters and $\tilde{P} = (\tilde{A}, \tilde{\sigma}, \tilde{\mu}, \tilde{B})^T$ the parameters estimated by msPE. Subscript n denotes a vector with each component being normalized to the range $[0, 1]$. Hence, the application of msPE on each signal results in 256 parameter vector pairs

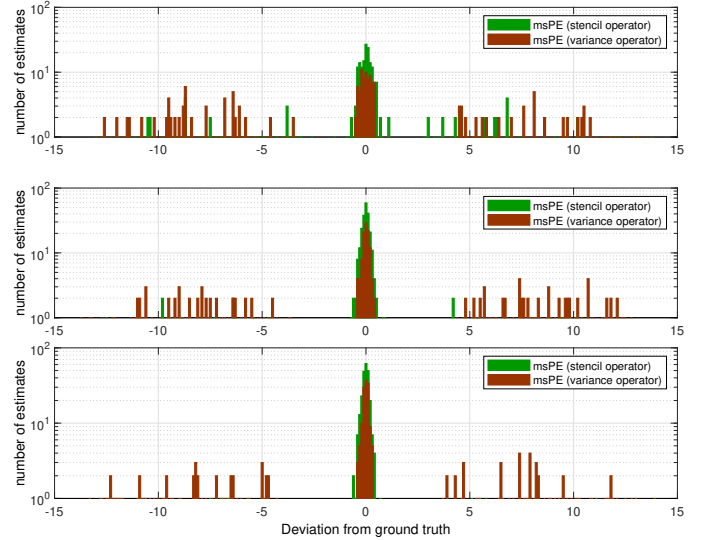


Fig. 5. These frequency distributions show the deviation of the estimated parameter μ from the ground truth. The "stencil operator" of this work is compared to the "variance operator" of previous work at selected SNR levels of -16 dB (top), -12 dB (center) and -10 dB (bottom).

$(P, \tilde{P})$ for a single line classifier. We create boxplots to compare the error distributions of both msPE configurations. Additionally, we compute frequency distributions (histograms) to compare the accuracy in estimating $\tilde{\mu}$.

III. RESULTS

Fig. 4 illustrates the distribution of error E [%] in form of boxplots, comparing msPE with the "stencil operator" (green) and the "variance operator" (red). For signals with rather high SNR, ranging from 0 dB to 50 dB, results are quite similar. At -10 dB a turning point is reached with the "stencil operator" showing a median error of 22% while the "variance operator" shows an error of 59% . In [4] we concluded that at this point a natural limit of the usefulness of the msPE is reached. For -12 dB results are similar with the median of the "variance operator" being 132% and the "stencil operator" showing 28% . At -16 dB the limit of the usefulness of the "stencil operator" is reached with the median E [%] being larger than 50% .

Results in Fig. 4 are errors averaged over all estimated parameters. To gain more insight into the improvement due to the "stencil operator", we analyze histograms showing the error of parameter μ only. This parameter directly depends on the correct classification of the WT^1 zero-crossing line.

Fig. 5 shows histograms for the error of μ for three SNR levels. At -10 dB both operators perform similarly and nearly always detect the correct WT^1 zero-crossing line. At -12 dB the "variance operator" is prone to errors and makes a wrong line classification in many cases, selecting the correct line in only up to 11% while the "stencil operator" selects the correct line in 23% . Fig. 6 shows an example of both msPE variants being applied to a Gaussian with a SNR of -16 dB.

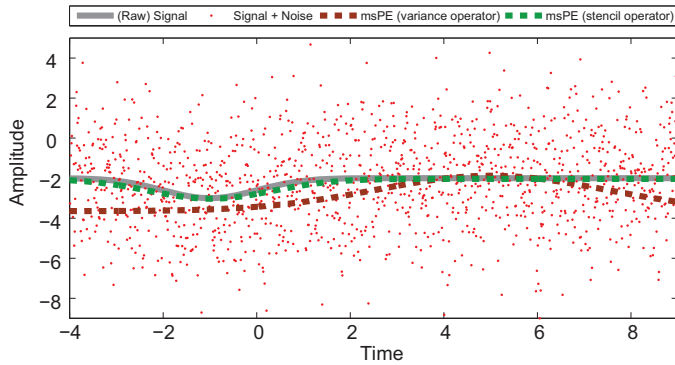


Fig. 6. Gaussian signal (gray curve; $A = -1$, $\sigma = 1$, $\mu = -1$, $B = -2$) with additive white (Gaussian) noise at -16 dB (red dots) and signals estimated by msPE with "variance operator" (brown curve) and "stencil operator" (green curve).

At -18 dB both operators fail in most cases but the results of the "stencil operator" are still outperforming the "variance operator".

IV. DISCUSSION AND CONCLUSION

The multiscale parameter estimation framework (msPE) [4] offers, for each model parameter to be estimated, a scale dependent algebraic equation (5)-(8). For μ and σ the respective parameter is expressed in terms of variables corresponding to zero-crossings of WT^1 and WT^2 . Here, "corresponding" refers to the mathematical relationship between a Gaussian's extrema, resp. inflection points and the zero-crossings in WT^1 , resp. WT^2 . Given the algebraic equations, the problem of estimating the model parameters of a noisy signal is reduced to the problem of assigning numerically obtained zero-crossings to the equation's corresponding variables $u0^1$ and $u0^2$. This in turn highly depends on correctly classifying zero-crossing lines.

In certain applications, where signal domain knowledge is available heuristics can be incorporated to improve line classification. In [10] we proposed a line classifier specifically customized to electrocardiography (ECG) signals but this method is use-case specific and can not be generalized to other applications.

In this work we propose a new zero-crossing line classifier for the general case of parameter estimation without prior knowledge. We evaluated our method using synthetic

Gaussian signals disturbed by normally-distributed noise. A method's robustness can be assessed from the underlying error distribution and reveals the limits of the existing line classifier ("variance operator") to lie at roughly 0 dB (Fig. 4). In other words the existing method is limited to situations, where signal and noise energy is roughly balanced. In contrast, the proposed "stencil operator" pushes this limit out to -12 dB, thus to scenarios where there is significantly more noise energy than signal energy (Fig. 6). The method represents an elegant and computationally efficient solution, since it only considers the CWT coefficients at the locations of the corresponding zero-crossing lines.

REFERENCES

- [1] B.-U. Kohler, C. Hennig, and R. Orglmeister, "The principles of software QRS detection," *IEEE Engineering in Medicine and Biology Magazine*, vol. 21, no. 1, pp. 42–57, 2002. [Online]. Available: <https://doi.org/10.1109/51.993193>
- [2] C. Yang, Z. He, and W. Yu, "Comparison of public peak detection algorithms for MALDI mass spectrometry data analysis," *BMC Bioinformatics*, vol. 10, no. 1, pp. 1, jan 2009. [Online]. Available: <https://doi.org/10.1186/1471-2105-10-4>
- [3] M. E. Mann and J. M. Lees, "Robust estimation of background noise and signal detection in climatic time series," *Climatic Change*, vol. 33, no. 3, pp. 409–445, jul 1996. [Online]. Available: <https://doi.org/10.1007/bf00142586>
- [4] M. Kukuk and N. Spicher, "Parameter Estimation Based on Scale-Dependent Algebraic Expressions and Scale-Space Fitting," *IEEE Transactions on Signal Processing*, vol. 67, no. 6, pp. 1431–1446, mar 2019. [Online]. Available: <https://doi.org/10.1109/tsp.2018.2887190>
- [5] H. Guo, "A simple algorithm for fitting a gaussian function [DSP tips and tricks]," *IEEE Signal Processing Magazine*, vol. 28, no. 5, pp. 134–137, sep 2011. [Online]. Available: <https://doi.org/10.1109/msp.2011.941846>
- [6] R. A. Caruana, R. B. Searle, T. Heller, and S. I. Shupack, "Fast algorithm for the resolution of spectra," *Analytical Chemistry*, vol. 58, no. 6, pp. 1162–1167, may 1986. [Online]. Available: <https://doi.org/10.1021/ac00297a041>
- [7] E. K. Roonizi, "A new algorithm for fitting a gaussian function riding on the polynomial background," *IEEE Signal Processing Letters*, vol. 20, no. 11, pp. 1062–1065, nov 2013. [Online]. Available: <https://doi.org/10.1109/lsp.2013.2280577>
- [8] S. Mallat and S. Zhong, "Characterization of signals from multiscale edges," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 14, no. 7, pp. 710–732, jul 1992. [Online]. Available: <https://doi.org/10.1109/34.142909>
- [9] A. L. Yuille and T. A. Poggio, "Scaling Theorems for Zero Crossings," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 1986. [Online]. Available: <https://doi.org/10.1109/TPAMI.1986.4767748>
- [10] N. Spicher and M. Kukuk, "Delineation of Electrocardiograms Using Multiscale Parameter Estimation," *IEEE Journal of Biomedical and Health Informatics*, vol. 24, no. 8, pp. 2216–2229, aug 2020. [Online]. Available: <https://doi.org/10.1109/jbhi.2019.2963786>